Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC I Chemistry Examination May 2017

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC-I) Chemistry. It contains comments on candidates' responses to the 2017 HSSC-I Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Comments

Overall, candidates performed well in this exam. In Constructed Response Questions (CRQs), they showed better understanding of the concepts such as calculation of mole ratios, Bohr's atomic theory, electronic configuration, chemical bonding/ shapes of molecules, and properties of liquids. However, a little more focus and efforts are required towards the concepts of limiting and excess reactants, Le-Chatelier's principle, extent of a chemical reaction, reverse osmosis and osmotic pressure.

In extended response questions (ERQs), candidates struggled to explain the Heisenberg's uncertainty principle and related concepts as compared to buffer solution and acid base chemistry, where the performance was better. Similarly, candidates displayed weak understanding while describing different types of solubility curves, but showed better progress to explain the lead storage battery.

Detailed Comments:

Constructed Response Questions (CRQs)

Question 1:

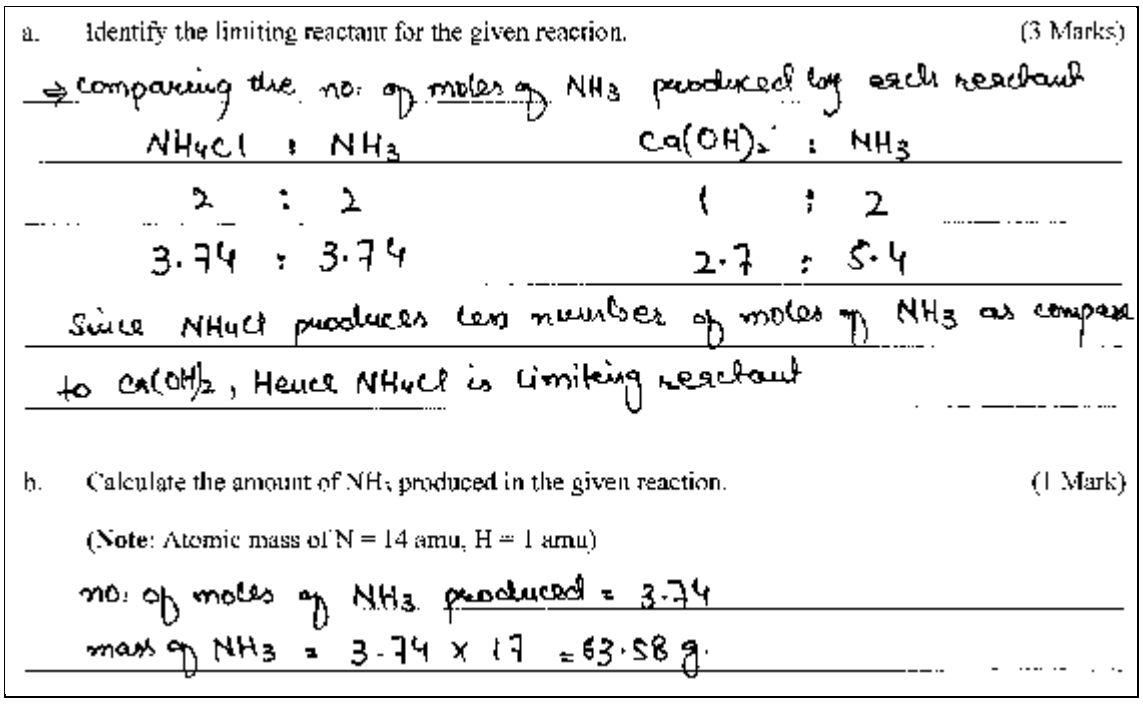
NH_3 is prepared by heating 3.74 moles of NH_4Cl and 2.7 moles of $Ca(OH)_2$.

$$2NH_4Cl_{(s)} + Ca(OH)_{2(s)} \longrightarrow CaCl_{2(s)} + 2NH_{3(g)} + 2H_2O_{(l)}$$

a. Identify the limiting reactant for the given reaction.
b. Calculate the amount of NH_3 produced in the given reaction.
(**Note**: Atomic mass of N = 14 amu, H = 1 amu)

Better responses exhibited that candidates compared moles of NH_4Cl and $Ca(OH)_2$ with NH_3 and found that the moles of NH_3 produced by NH_4Cl are lesser, which identified NH_4Cl as a limiting reagent. Few responses compared moles of NH_4Cl with $Ca(OH)_2$ to discover $Ca(OH)_2$ in excess and identified NH_4Cl as a limiting reagent. Furthermore, candidates correctly calculated the amount of NH_3 produced by multiplying the moles of NH_3 produced with its molecular weight.

Example:



a. Identify the limiting reactant for the given reaction. (3 Marks)

⇒ comparing the no. of moles of NH_3 produced by each reactant

NH_4Cl : NH_3	$Ca(OH)_2$: NH_3
2 : 2	1 : 2
3.74 : 3.74	2.7 : 5.4

Since NH_4Cl produces less number of moles of NH_3 as compared to $Ca(OH)_2$, Hence NH_4Cl is limiting reactant

b. Calculate the amount of NH_3 produced in the given reaction. (1 Mark)

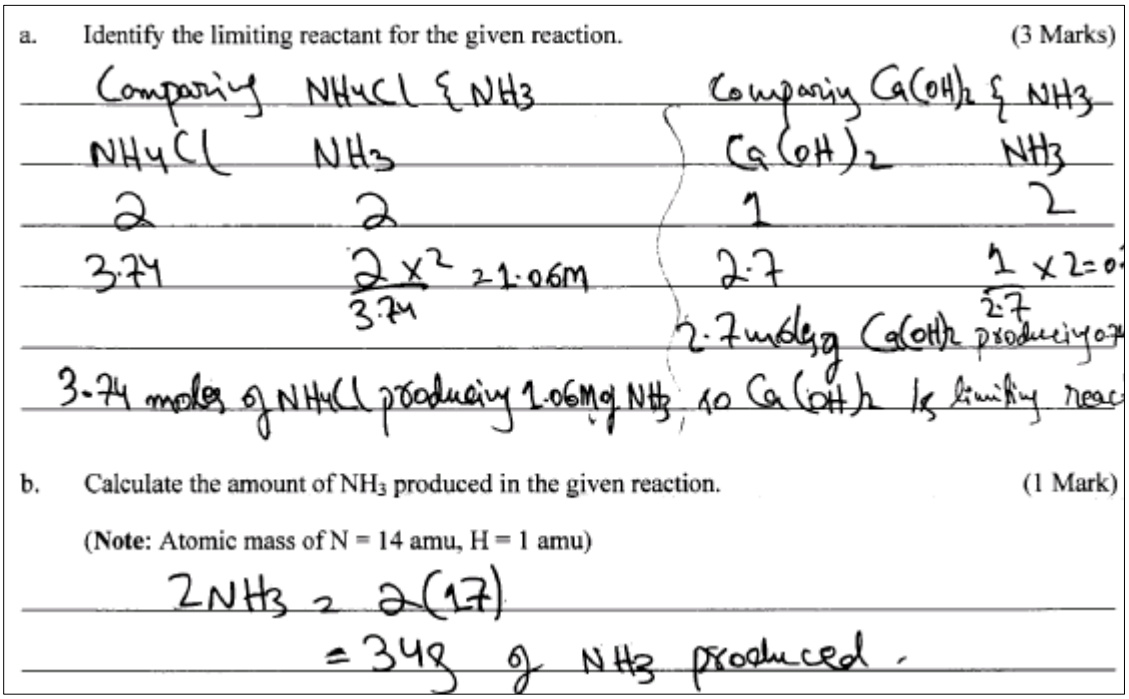
(Note: Atomic mass of N = 14 amu, H = 1 amu)

no. of moles of NH_3 produced = 3.74

mass of NH_3 = 3.74 × 17 = 63.58 g.

Weaker responses revealed that candidates compared moles of NH_4Cl and $Ca(OH)_2$ with NH_3 but were unable to find correct moles of NH_3 and wrongly identified $Ca(OH)_2$ as a limiting reagent. Furthermore, using the incorrect moles of NH_3 they calculated wrong amount of NH_3 produced in a reaction.

Example:


a. Identify the limiting reactant for the given reaction. (3 Marks)

Comparing NH_4Cl & NH_3 | Comparing $Ca(OH)_2$ & NH_3

NH_4Cl	NH_3	$Ca(OH)_2$	NH_3
2	2	1	2
3.74	$\frac{2 \times 2}{3.74} = 1.06M$	2.7	$\frac{1 \times 2}{2.7} = 0.$

2.7 moles of $Ca(OH)_2$ producing 0.

3.74 moles of NH_4Cl producing 1.06M of NH_3, so $Ca(OH)_2$ is limiting reac

b. Calculate the amount of NH_3 produced in the given reaction. (1 Mark)

(**Note:** Atomic mass of N = 14 amu, H = 1 amu)

$2NH_3 = 2(17)$

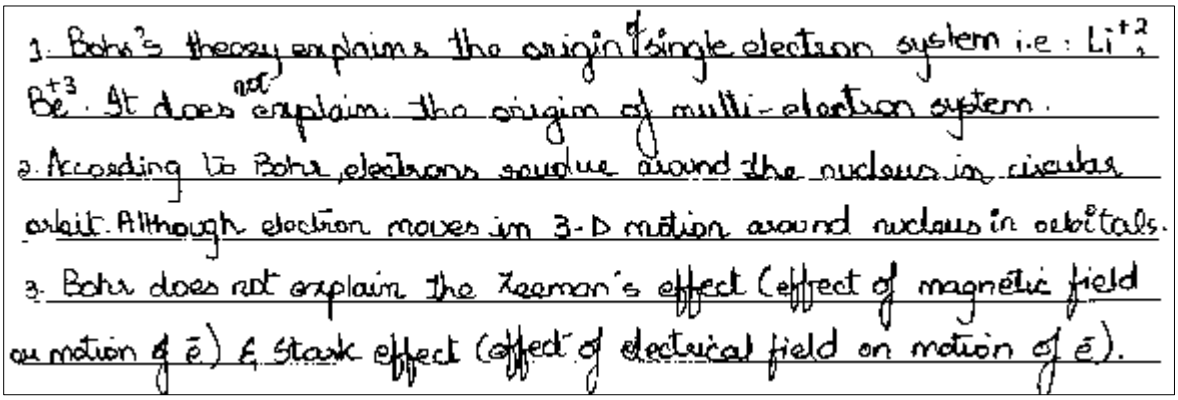
$= 34g$ of NH_3 produced.

Question 2a:

State any THREE defects of Bohr's atomic model.

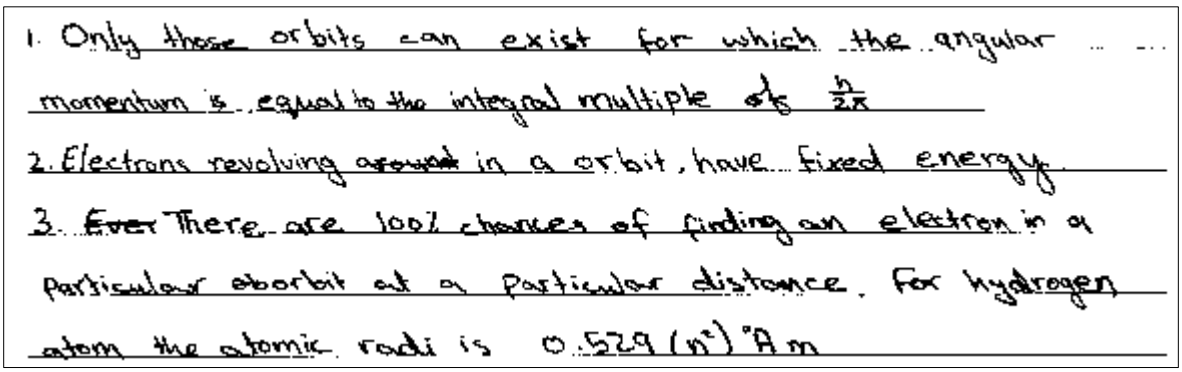
Better responses showed clear understanding of Bohr's atomic model and mentioned the defects associated with this model such as it does not explain either Zeeman or Stark effect, origin of the spectrum of multi-electrons system and motion of electron in three dimensional space.

Example:


1. Bohr's theory explains the origin of single electron system i.e: Li^{+2}, Be^{+3}. It does not explain the origin of multi-electron system.
2. According to Bohr electrons revolve around the nucleus in circular orbit. Although electron moves in 3-D motion around nucleus in orbitals.
3. Bohr does not explain the Zeeman's effect (effect of magnetic field on motion of $\bar{e}$) & Stark effect (effect of electrical field on motion of $\bar{e}$).

Weaker responses showed poor knowledge of the concept of Bohr's atomic model. Instead of writing the defects of Bohr's atomic model, candidates wrote a few points related to Heisenberg's uncertainty principle. These responses discussed about angular momentum, energy of electron, atomic radius and probability of finding electron in an atom.

Example:

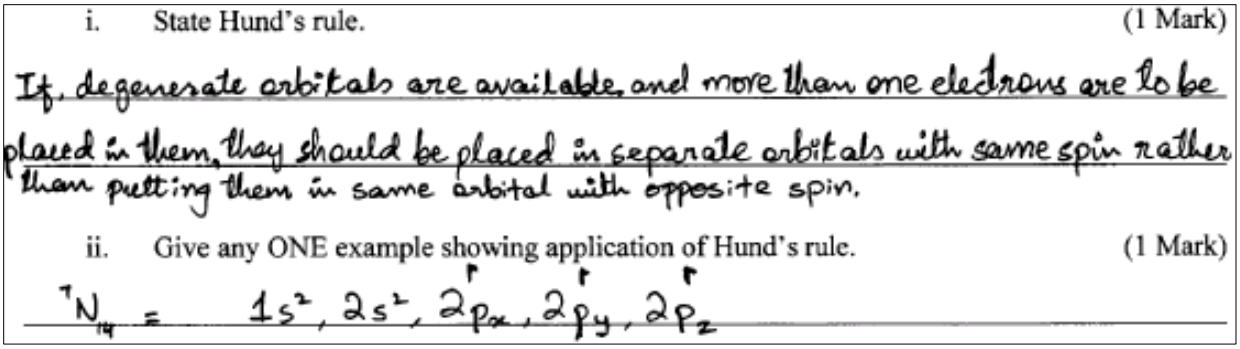

1. Only those orbits can exist for which the angular momentum is equal to the integral multiple of $\frac{h}{2\pi}$
2. Electrons revolving in a orbit, have fixed energy.
3. Ever There are 100% chances of finding an electron in a particular eborbit at a particular distance. For hydrogen atom the atomic radi is $0.529 (n^2)$ °A m

Question 2b:

i. State Hund's rule.
ii. Give any ONE example showing application of Hund's rule.

Better responses stated Hund's rule clearly using the keywords such as availability of degenerate orbitals and placement of electrons with same/ opposite spin. Most of these responses used carbon or nitrogen atom as an example to show the distribution of electrons in subshells and to explain the Hund's rule.

Example:


i. State Hund's rule. (1 Mark)

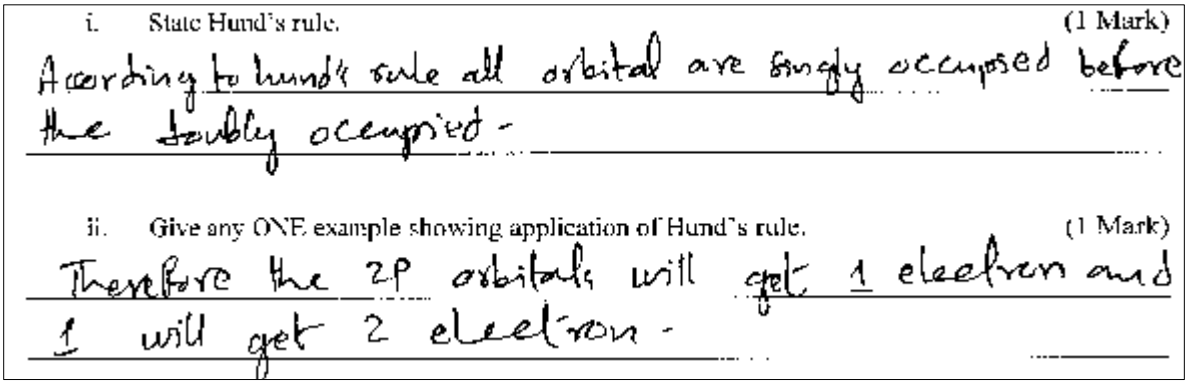
If, degenerate orbitals are available, and more than one electrons are to be placed in them, they should be placed in separate orbitals with same spin rather than putting them in same orbital with opposite spin.

ii. Give any ONE example showing application of Hund's rule. (1 Mark)

$^{7}N_{14} = 1s^2, 2s^2, 2p_x, 2p_y, 2p_z$

Weaker responses failed to correctly define Hund's rule. Most of these responses discussed the distribution of electrons in subshells with increasing energy of orbitals, while few of these responses mentioned the distribution of electrons in separate degenerate orbitals, but did not discuss the spin of electrons.

Example:


i. State Hund's rule. (1 Mark)

According to hund's rule all orbital are singly occupied before the doubly occupied -

ii. Give any ONE example showing application of Hund's rule. (1 Mark)

Therefore the 2P orbitals will get 1 electron and 1 will get 2 electron -

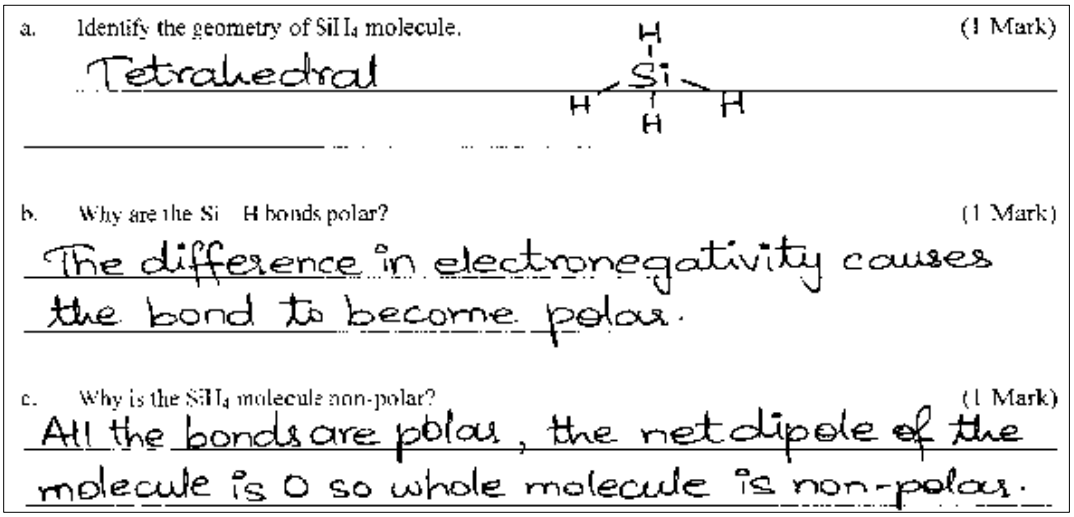
Question 3:

SiH_4 is a non-polar covalent molecule with polar bonds.

a. Identify the geometry of SiH_4 molecule.
b. Why are the bonds polar?
c. Why is the SiH_4 molecule non-polar?

Better responses depicted in-depth understanding and relation between polarity, symmetry and dipole moment. High-scoring candidates clearly recognised the geometry of SiH_4 molecule as tetrahedral, and discussed the difference of electronegativity between Si and H atom, which makes Si-H bond polar. They provided correct justification about the non-polar nature of SiH_4 molecule in terms of net dipole moment. Some of these responses discussed the symmetry in molecules which is responsible for the non-polar behaviour of SiH_4 molecule.

Example:


a. Identify the geometry of SiH_4 molecule. (1 Mark)

Tetrahedral

b. Why are the Si – H bonds polar? (1 Mark)

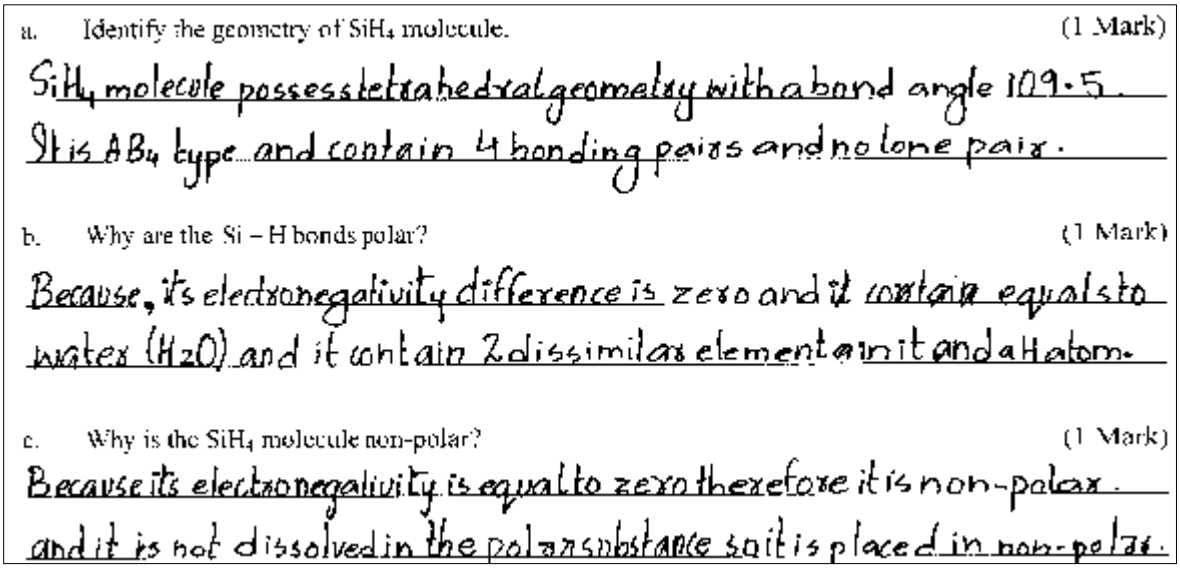
The difference in electronegativity causes the bond to become polar.

c. Why is the SiH_4 molecule non-polar? (1 Mark)

All the bonds are polar, the net dipole of the molecule is 0 so whole molecule is non-polar.

Weaker responses recognised the geometry of SiH_4 molecule as tetrahedral, however, correct reason behind the polarity in Si-H bond and non-polar nature of SiH_4 molecule was not provided. Instead of dipole moment, candidates considered electronegativity difference between Si and H atoms as zero and responsible for non-polar nature of SiH_4 molecule.

Example:



a. Identify the geometry of SiH_4 molecule. (1 Mark)

SiH_4 molecule possess tetrahedral geometry with a bond angle 109.5. It is AB_4 type and contain 4 bonding pairs and no lone pair.

b. Why are the Si – H bonds polar? (1 Mark)

Because, its electronegativity difference is zero and it contain equals to water (H_2O) and it contain 2 dissimilar element in it and a H atom.

c. Why is the SiH_4 molecule non-polar? (1 Mark)

Because its electronegativity is equal to zero therefore it is non-polar. and it is not dissolved in the polar substance so it is placed in non-polar.

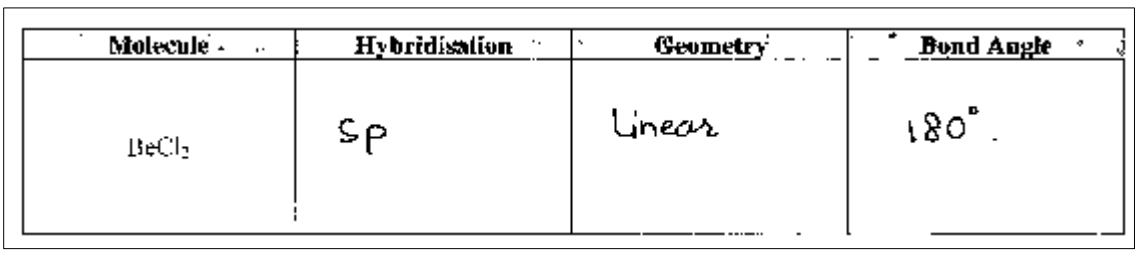
Question 4:

Identify the type of hybridisation, geometry and bond angle between hybrid orbitals of the molecule in the given table.

Molecule	Hybridisation	Geometry	Bond Angle
$BeCl_2$			

Better responses demonstrated good understanding of the shapes of simple molecules and their orbital hybridization. These responses interpreted the structure of $BeCl_2$ molecule clearly and completed the table provided with correct type of hybridisation, geometry and bond angle in $BeCl_2$ molecule.

Example:



Molecule	Hybridisation	Geometry	Bond Angle
$BeCl_2$	sp	Linear	180°.

Weaker responses exhibited that candidates failed to recognise $BeCl_2$ molecule. Most of these responses identified $BeCl_2$ as tetrahedral molecule and suggested sp^3 hybridisation and 109.5^o bond angle for the same. Similarly, fewer responses selected sp^2 hybridisation and trigonal geometry with respective bond angle.

Example:

Molecule	Hybridisation	Geometry	Bond Angle
$BeCl_2$	sp^3	tetrahydral	109.5°

Question 5:

Calculate the mass of 10 dm^3 of oxygen (O2) gas at 80°C and 3000 mm Hg pressure.
(**Note**: The value of R = 0.0821 dm^3 atm K^{-1} mol^{-1} and atomic mass of O = 16 amu)

Better responses depicted stepwise procedure to calculate the mass of oxygen. Initially, they showed unit conversion and changed temperature from Celsius to Kelvin scale and pressure from mm Hg to atm. Then, they used appropriate formula and substituted values correctly to calculate the mass of oxygen in gram.

Example:

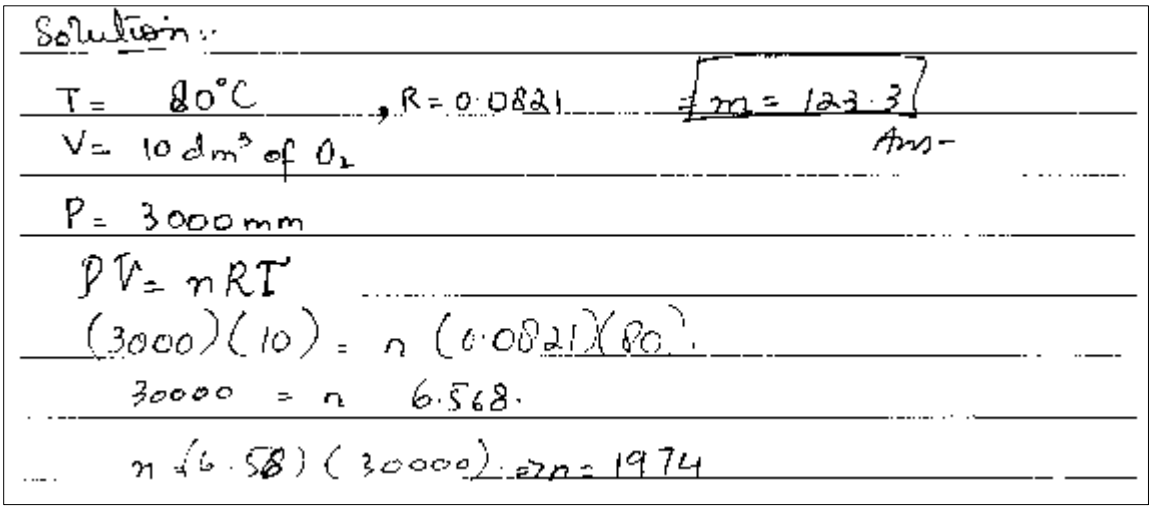
Data:-
V = 10dm³
T = 80°C + 273 = 353K
P = 3000mm of Hg / 760 = 3.94atm
R = 0.0821 dm³ atm K⁻¹ mol⁻¹
Molar Mass = 32 g/mol
Mass = ?

Solution:-
PV = Mass / Molar Mass RT
Mass = PV x Mr / RT
Mass = (3.94)(10)(32) / (0.0821)(353)
Mass = 43.58gm

∴ Mass is 43.58gm or 44gm

Weaker responses struggled to calculate the correct mass of oxygen. These responses displayed careless errors such as temperature or pressure conversion was not demonstrated. Some of these responses apply inappropriate formula or incorrect substitution of values. However, some of these responses used correct formula and substitution of values, but calculated wrong answer with no unit mentioned in the final answer.

Example:



Question 6:

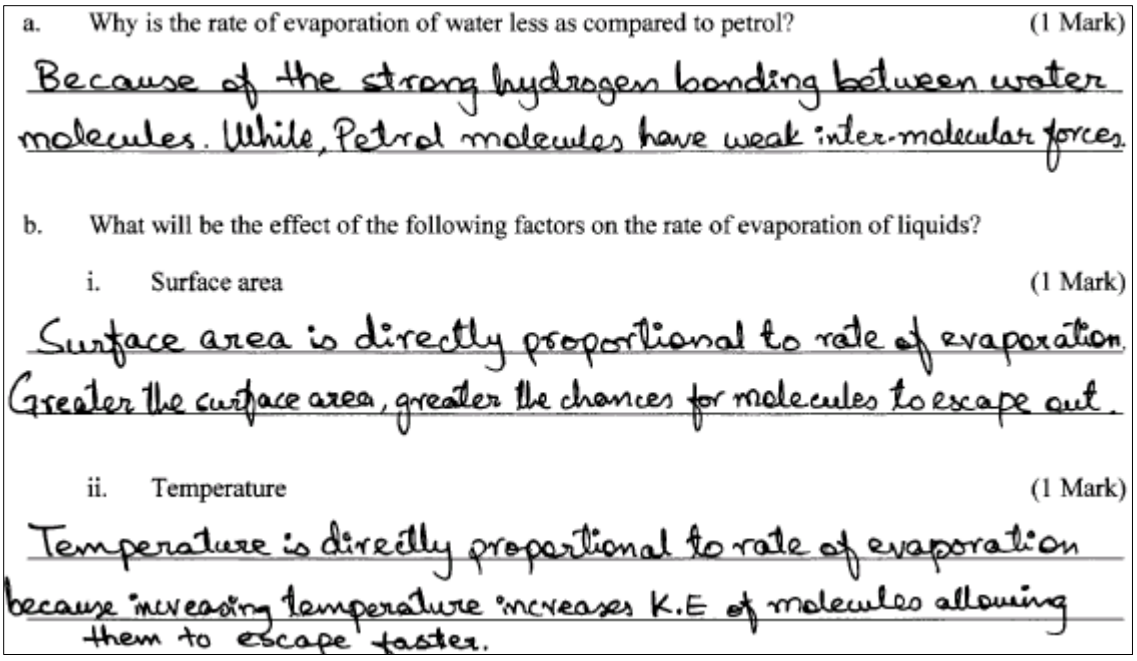
a. Why is the rate of evaporation of water less as compared to petrol?

b. What will be the effect of the following factors on the rate of evaporation of liquids?

   i. Surface area
   ii. Temperature

Better responses gave correct justification about why the rate of evaporation of water is lesser as compared to petrol. Candidates explained that water possesses strong intermolecular forces in terms of hydrogen bonding, which is much stronger force of attraction than the weak dispersion forces existing in petrol molecule. In part b, these responses correctly displayed the effect of surface area and temperature on the rate of evaporation of liquids. They specified direct relationship between surface area or temperature with rate of evaporation of liquids, and mentioned that as the temperature or surface area increases the rate of evaporation of liquids increases or vice versa. Some of these responses also provided additional information and explained the reason behind why the surface area and temperature has direct relation with rate of evaporation.

Example:



a. Why is the rate of evaporation of water less as compared to petrol? (1 Mark)

Because of the strong hydrogen bonding between water molecules. While, Petrol molecules have weak inter-molecular forces.

b. What will be the effect of the following factors on the rate of evaporation of liquids?

i. Surface area (1 Mark)

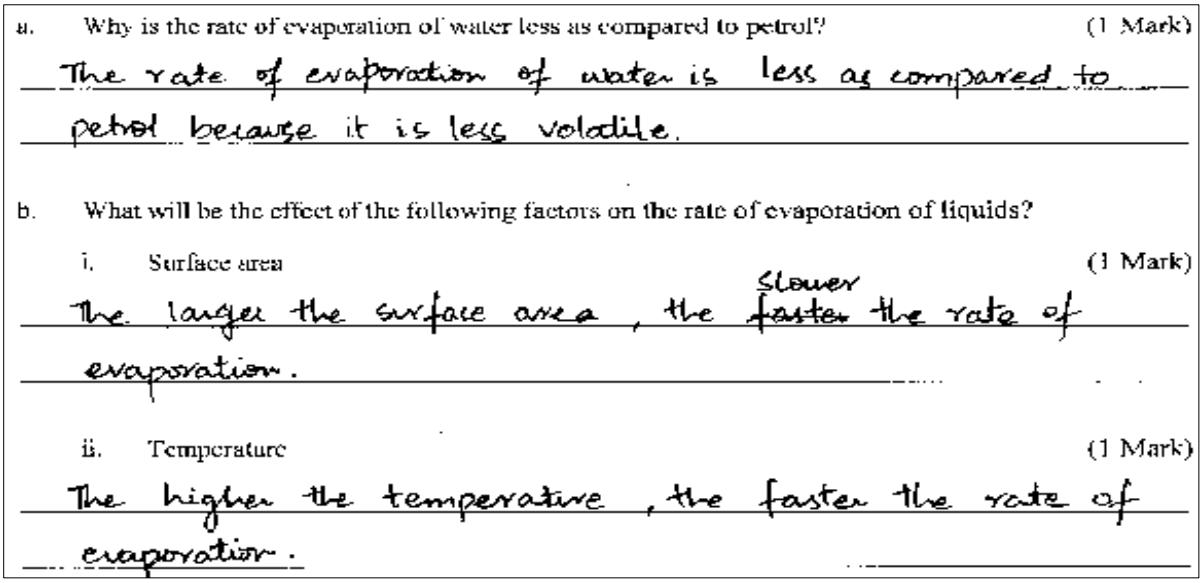
Surface area is directly proportional to rate of evaporation. Greater the surface area, greater the chances for molecules to escape out.

ii. Temperature (1 Mark)

Temperature is directly proportional to rate of evaporation because increasing temperature increases K.E of molecules allowing them to escape faster.

Weaker responses displayed wrong judgment while explaining the reason behind lesser rate of evaporation of water as compared to petrol. These responses mentioned variety of factors such as high density, low viscosity or low volatility of water molecule, but failed to discuss the intermolecular forces that exist in water or petrol molecule. In part b, most of the low scoring candidates showed wrong interpretation that the rate of evaporation decreases with increase in surface area of liquids. However, some of them mentioned the direct relation of temperature with rate of evaporation. Few of these responses only stated that rate of evaporation increases in both cases, but did not clarify the exact relation about whether it increases with increase in surface area or temperature or it increases with decrease in surface area or temperature.

Example:



a. Why is the rate of evaporation of water less as compared to petrol? (1 Mark)

The rate of evaporation of water is less as compared to petrol because it is less volatile.

b. What will be the effect of the following factors on the rate of evaporation of liquids?

i. Surface area (1 Mark)

The larger the surface area, the ~~faster~~ slower the rate of evaporation.

ii. Temperature (1 Mark)

The higher the temperature, the faster the rate of evaporation.

Question 7:

a. Define reversible reaction and give ONE example of it.

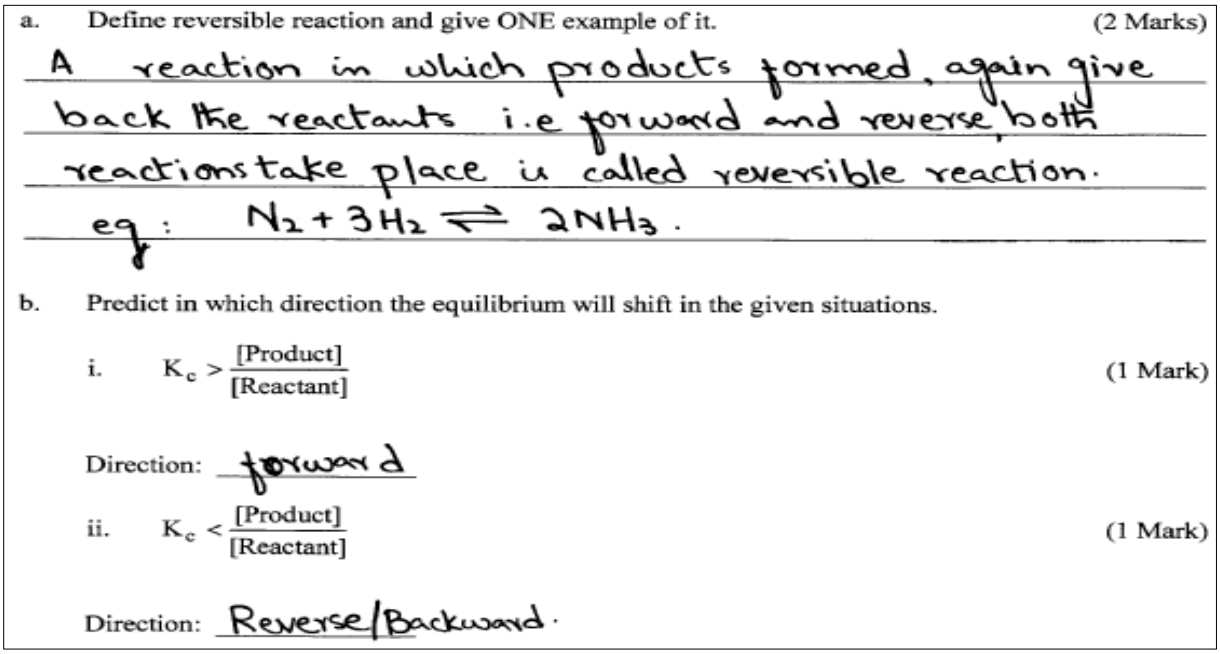
b. Predict in which direction the equilibrium will shift in the given situations.

i. $K_c > \frac{[Product]}{[Reactant]}$

ii. $K_c < \frac{[Product]}{[Reactant]}$

Better responses provided correct definition of reversible reaction highlighting the information such as product once formed can react to form reactants or reactions take place both in the forward and backward directions. Moreover, candidates wrote correct example representing reactants and products in a chemical equation in relation to the definition of reversible reaction. In part b, these responses indicated the correct direction of equilibrium under the given situation. In subpart i, where the K_c is larger than the ratio of product and reactant, candidates mentioned the direction as forward. Similarly, in subpart ii, where the K_c is smaller than the ratio of product and reactant, candidates mentioned the direction as backward. Alternatively, some of these responses used the words 'left or right' instead of forward or backward'.

Example:


a. Define reversible reaction and give ONE example of it. (2 Marks)

A reaction in which products formed, again give back the reactants i.e forward and reverse both reactions take place is called reversible reaction.

eg: $N_2 + 3H_2 \rightleftharpoons 2NH_3$.

b. Predict in which direction the equilibrium will shift in the given situations.

i. $K_c > \frac{[Product]}{[Reactant]}$ (1 Mark)

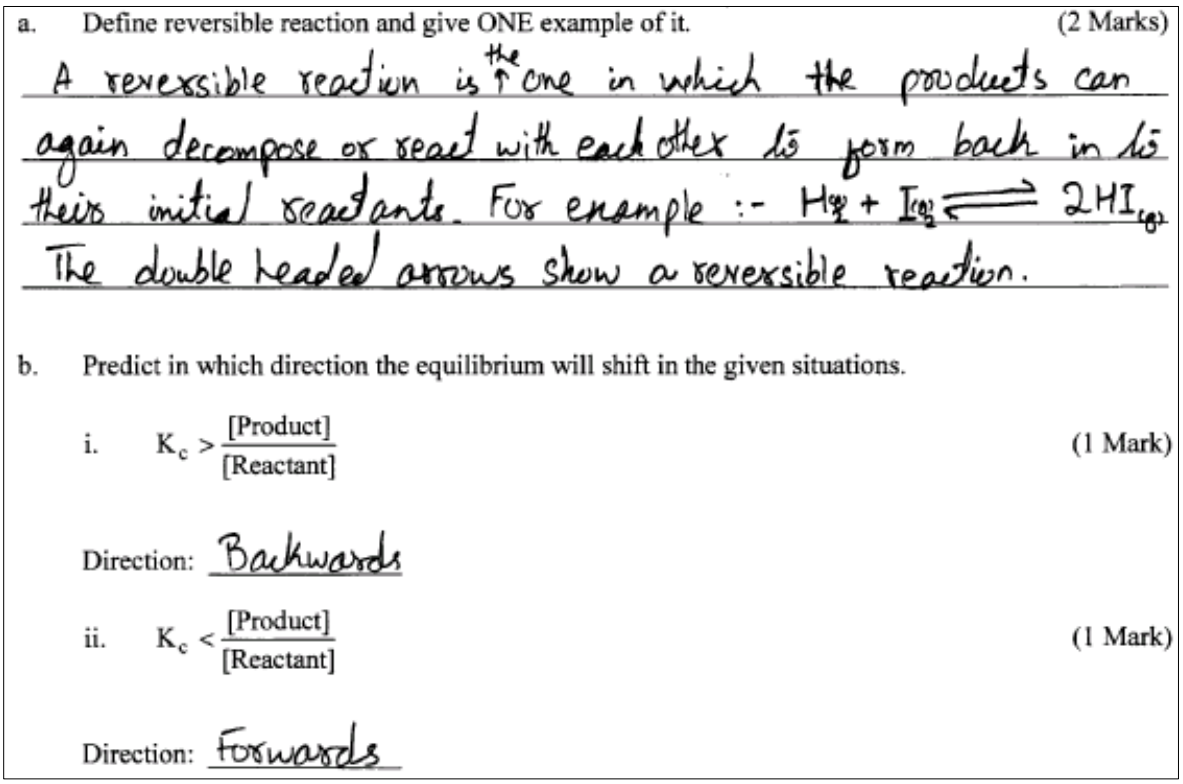
Direction: forward

ii. $K_c < \frac{[Product]}{[Reactant]}$ (1 Mark)

Direction: Reverse/Backward.

Weaker responses showed lack of knowledge about the concept of chemical equilibrium and reversible reaction. Candidates were not able to provide correct definition of reversible reaction. However, some of the weak responses provided correct definition of reversible reaction, but they failed to give a relevant example. In subpart i and ii of part b, these responses deduced the direction of equilibrium wrongly. Instead of 'forward', they identified the direction of equilibrium as 'backward' and vice versa.

Example:



a. Define reversible reaction and give ONE example of it. (2 Marks)

A reversible reaction is the one in which the products can again decompose or react with each other to form back in to their initial reactants. For example :- $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$ The double headed arrows show a reversible reaction.

b. Predict in which direction the equilibrium will shift in the given situations.

i. $K_c > \frac{[Product]}{[Reactant]}$ (1 Mark)

Direction: Backwards

ii. $K_c < \frac{[Product]}{[Reactant]}$ (1 Mark)

Direction: Forwards

Question 8:

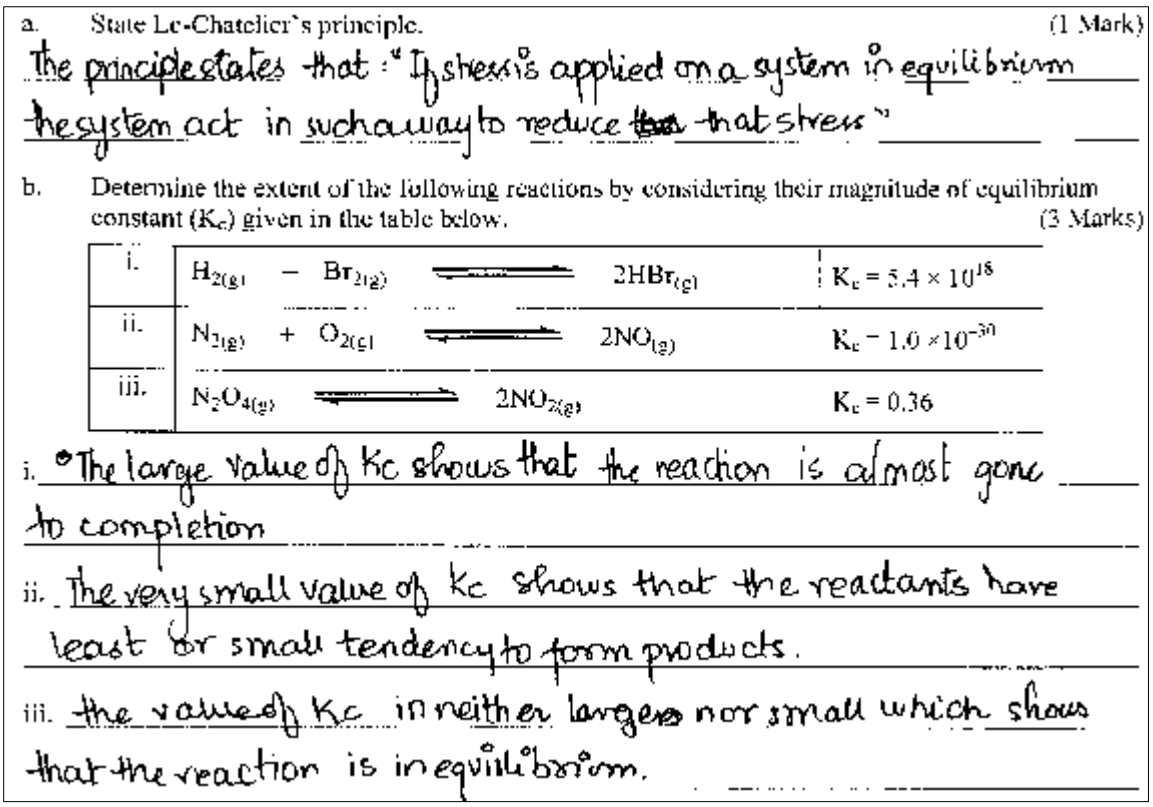
a. State Le-Chatelier's principle.

b. Determine the extent of the following reactions by considering their magnitude of equilibrium constant (K_c) given in the table below.

i.	$H_{2(g)} + Br_{2(g)} \rightleftharpoons 2HBr_{(g)}$	$K_c = 5.4 \times 10^{18}$
ii.	$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$	$K_c = 1.0 \times 10^{-30}$
iii.	$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$	$K_c = 0.36$

Better responses clearly stated the Le-Chatelier's principle mentioning keywords like 'stress is applied to a system at equilibrium'. In part b, these responses correctly interpreted the extent of each reaction by considering the magnitude of given K_c values. In subpart i, these responses mentioned that the reaction goes virtually/ almost to completion as the value of K_c is too large. On the other hand, in subpart ii, candidates demonstrated that the reaction does not proceed appreciably in the forward direction or proceeds in the backward direction as the value of K_c is too small. Finally, in subpart iii, these responses indicated that neither forward nor reverse reaction goes to completion because of the moderate K_c value.

Example:


a. State Le-Chatelier's principle. (1 Mark)

the principle states that: "If stress is applied on a system in equilibrium the system act in such a way to reduce that stress"

b. Determine the extent of the following reactions by considering their magnitude of equilibrium constant (K_c) given in the table below. (3 Marks)

i.	$H_{2(g)} - Br_{2(g)} \rightleftharpoons 2HBr_{(g)}$	$K_c = 5.4 \times 10^{18}$
ii.	$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$	$K_c = 1.0 \times 10^{-30}$
iii.	$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$	$K_c = 0.36$

i. The large value of Kc shows that the reaction is almost gone to completion

ii. The very small value of Kc shows that the reactants have least or small tendency to form products.

iii. the value of Kc in neither larger nor small which shows that the reaction is in equilibrium.

Weaker responses provided wrong information about the Le-Chatelier's principle. Some of these responses gave incomplete details or inappropriate definition of Le-Chatelier's principle. Candidates did not mention that the 'system is at equilibrium'. Similarly, instead of writing 'the system acts in such a way to nullify the effect of that stress', candidates wrote 'equilibrium will shift in opposite direction'. In subpart i, ii, iii of part b, most of the weak responses gave irrelevant calculations. Moreover, most candidates tried to deduce the equilibrium expression for the given reactions instead of considering the value of K_c to interpret the extent of reactions.

Example:

a. State Le-Chatelier's principle. (1 Mark)

It states that if reaction is disturbed the equilibrium will shift to opposite direction

b. Determine the extent of the following reactions by considering their magnitude of equilibrium constant (K_c) given in the table below. (3 Marks)

i.	$H_{2(g)} + Br_{2(g)} \rightleftharpoons 2HBr_{(g)}$	$K_c = 5.4 \times 10^{18}$
ii.	$N_{2(g)} + O_{2(g)} \rightleftharpoons 2NO_{(g)}$	$K_c = 1.0 \times 10^{-30}$
iii.	$N_2O_{4(g)} \rightleftharpoons 2NO_{2(g)}$	$K_c = 0.36$

i. $5.4 \times 10^{18} = \frac{[HBr]^2}{[H_2][Br_2]}$

ii. $1.0 \times 10^{-30} = \frac{[NO]^2}{[N_2][O_2]}$

iii. $0.36 = \frac{[NO_2]^2}{[N_2O_4]}$

Question 9:

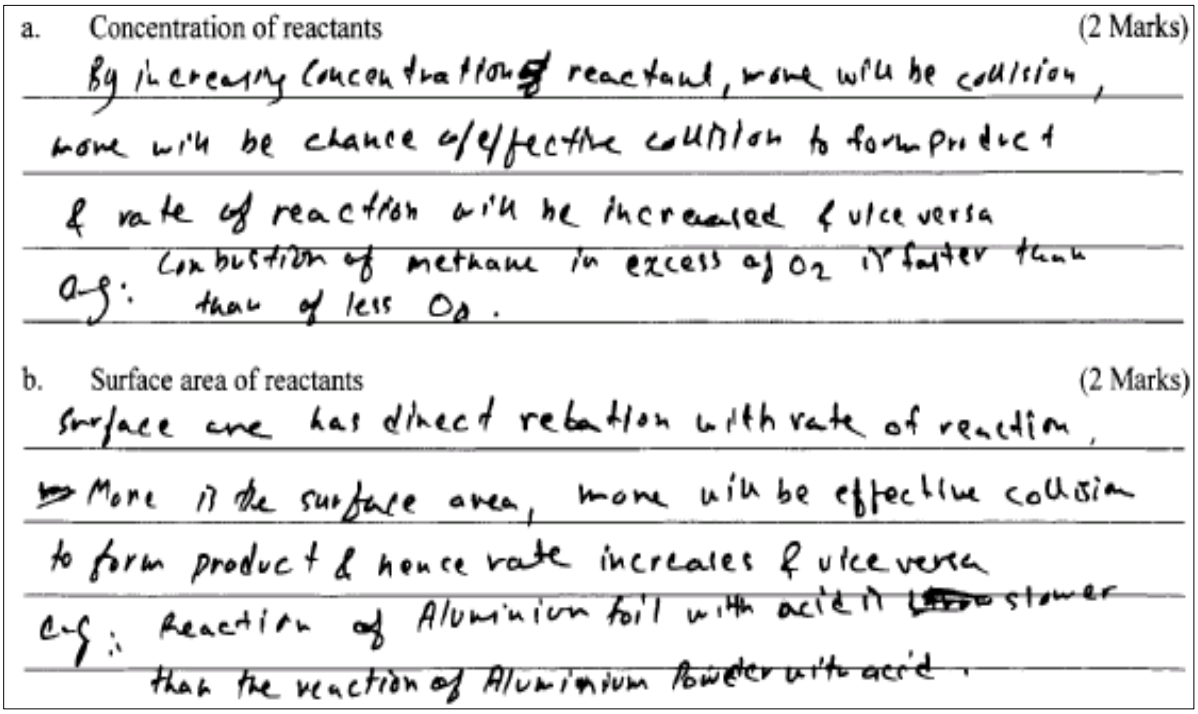
On the basis of collision theory, describe the effect of the following factors on the rate of a chemical reaction. Give ONE example in each case.

a. Concentration of reactants

b. Surface area of reactants

Better responses described the effect of concentration and surface area of reactants on the rate of a chemical reaction and specified relevant examples to support their description. Candidates described that an increase in the concentration of the reactants will result in more collision among molecules, which increases the rate of reaction. Candidates gave example of combustion reaction that occurs slowly in air, but occur more rapidly in pure oxygen. Similarly, they explained that as the surface area increases the possibilities of atoms and molecules of reactants to come in contact with each other increases; this enhances the rate of reaction. Powdered aluminium was mentioned as an example, which reacts rapidly with acid or bases compared to aluminium foil.

Example:



a. Concentration of reactants (2 Marks)

By increasing Concentration of reactant, more will be collision, more will be chance of effective collision to form product & rate of reaction will be increased & vice versa

eg: Combustion of methane in excess of O_2 is faster than that of less O_2.

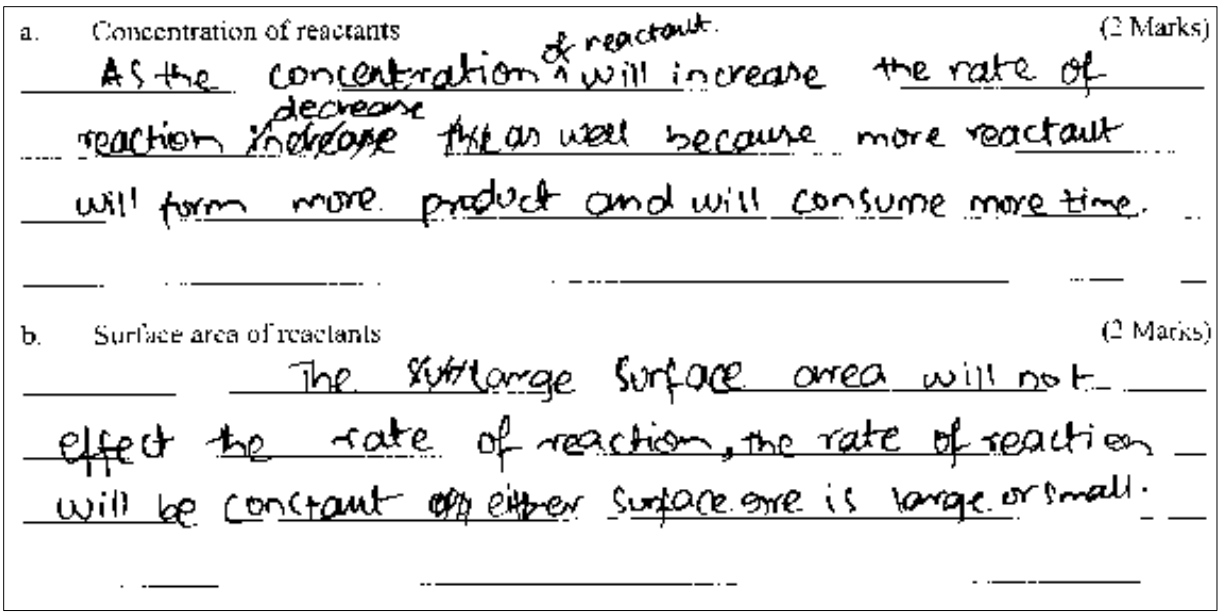
b. Surface area of reactants (2 Marks)

Surface area has direct relation with rate of reaction, More is the surface area, more will be effective collision to form product & hence rate increases & vice versa

eg: Reaction of Aluminium foil with acid is slower than the reaction of Aluminium Powder with acid.

Weaker responses showed wrong relationship between the concentration/ surface area of reactants with rate of reaction. These responses exhibited illogical explanation that as the concentration of reactants increases, the rate of reaction decreases because large number of molecules will take more time to react with each other. Similarly, a few other responses indicated that the rate of reaction is independent to the surface area of reactant. In addition, these responses were unable to provide any example to support their explanations.

Example:


a. Concentration of reactants (2 Marks)

As the concentration of reactant will increase the rate of reaction decrease/increase this as well because more reactant will form more product and will consume more time.

b. Surface area of reactants (2 Marks)

The ~~s~~/large surface area will not effect the rate of reaction, the rate of reaction will be constant of either surface are is large or small.

Question 10:

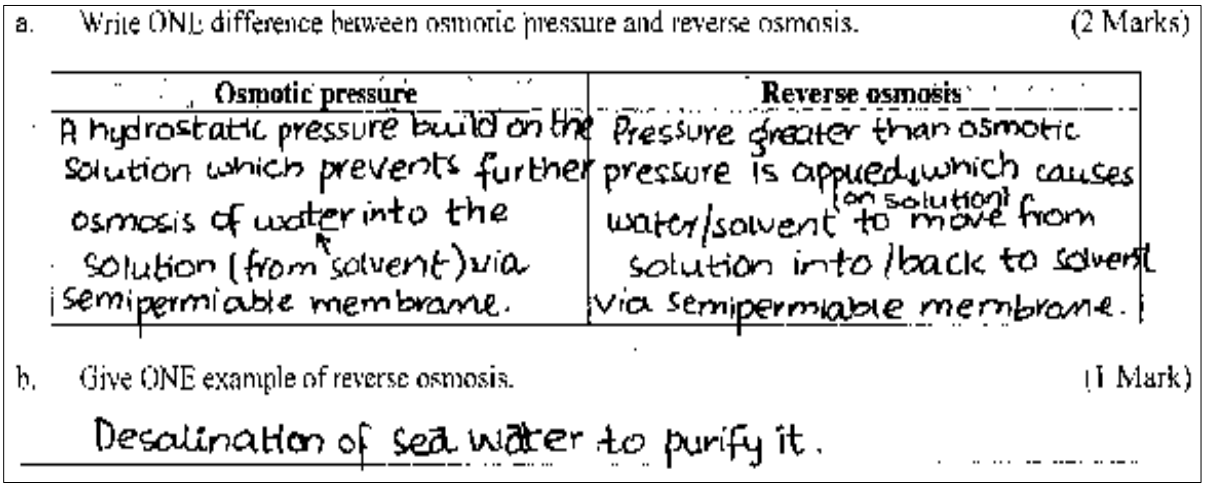
a. Write ONE difference between osmotic pressure and reverse osmosis.

Osmotic pressure	Reverse osmosis

b. Give ONE example of reverse osmosis.

Better responses depicted good understanding of osmosis and reverse osmosis and demonstrated correct difference between osmotic pressure and reverse osmosis. These responses explained osmotic pressure as the pressure applied to a pure solvent to prevent it from passing into a given solution through the process of osmosis. Candidates described reverse osmosis as a process by which a solvent passes through a porous membrane in the direction opposite to natural osmosis. Some of these responses mentioned that the magnitude of pressure applied during reverse osmosis is larger than the osmotic pressure. Furthermore, they represented desalination of sea water as an example of reverse osmosis.

Example:


a. Write ONE difference between osmotic pressure and reverse osmosis. (2 Marks)

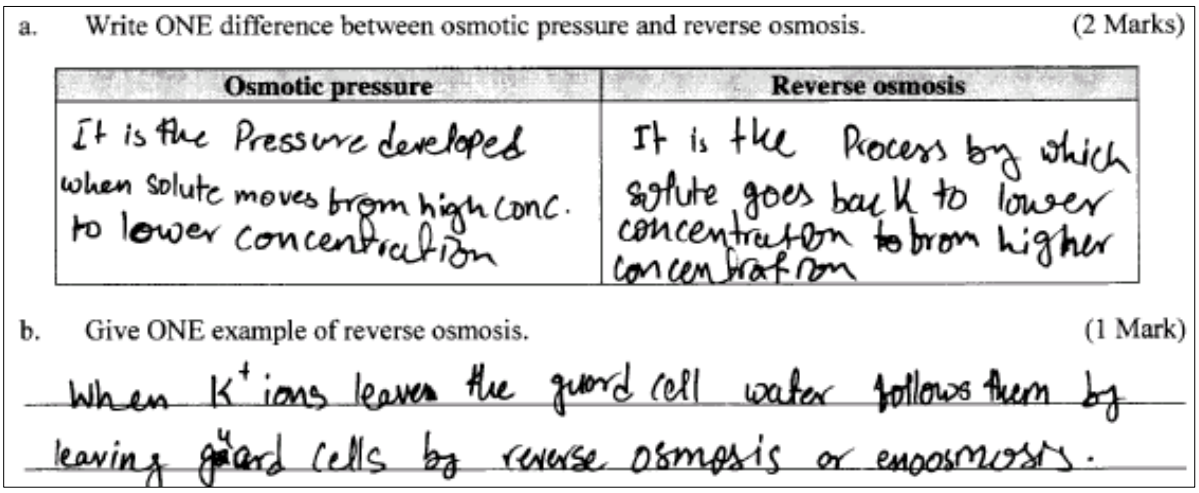
Osmotic pressure	Reverse osmosis
A hydrostatic pressure build on the solution which prevents further osmosis of water into the solution (from solvent) via semipermiable membrane.	Pressure greater than osmotic pressure is appued which causes water/solvent (on solution) to move from solution into/back to solvent via semipermiable membrane.

b. Give ONE example of reverse osmosis. (1 Mark)

Desalination of sea water to purify it.

Weaker responses provided wrong explanation of osmotic pressure and reverse osmosis. In some of these responses, candidates tried to explain the process of osmosis or indicated the direction of flow of pure solvent during osmosis, but did not discuss the osmotic pressure. At times, an incomplete explanation of reverse osmosis indicating only the direction of flow of solvent was given, however, external pressure was not mentioned. Moreover, these responses were unable to provide correct example of reverse osmosis.

Example:



a. Write ONE difference between osmotic pressure and reverse osmosis. (2 Marks)

Osmotic pressure	Reverse osmosis
It is the Pressure developed when solute moves from high conc. to lower concentration	It is the Process by which solute goes back to lower concentration to from higher concentration

b. Give ONE example of reverse osmosis. (1 Mark)

When K^+ ions leaves the guard cell water follows them by leaving guard cells by reverse osmosis or endosmosis.

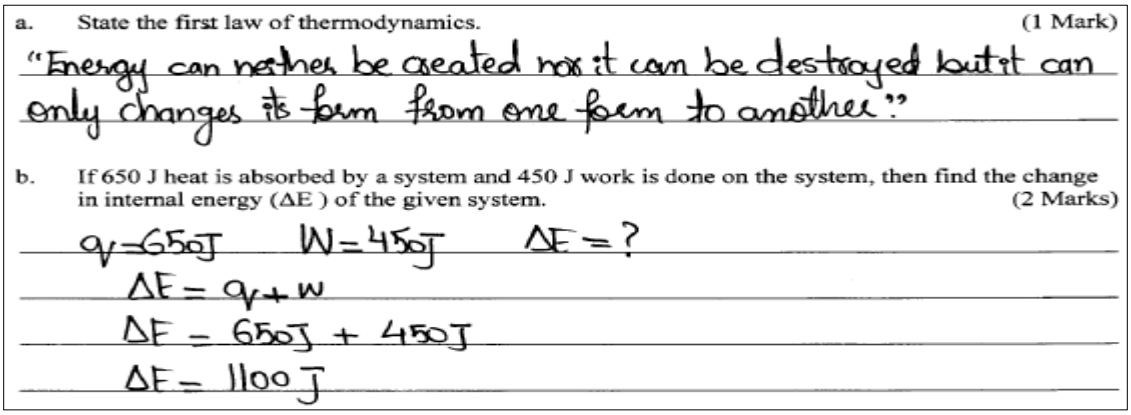
Question 11:

a. State the first law of thermodynamics.

b. If 650 J heat is absorbed by a system and 450 J work is done on the system, then find the change in internal energy (ΔE) of the given system.

Better responses clearly stated the first law of thermodynamics highlighting that energy can neither be created nor destroyed but it can be changed from one form to another. In part b, candidates analysed the given situation very well, i.e. work is done on the system, and displayed correct calculation to calculate the change in internal energy of the system using appropriate formula and substitution of values. These responses also showed correct unit in their final answer.

Example:


a. State the first law of thermodynamics. (1 Mark)

"Energy can neither be created nor it can be destroyed but it can only changes its form from one form to another."

b. If 650 J heat is absorbed by a system and 450 J work is done on the system, then find the change in internal energy (ΔE) of the given system. (2 Marks)

$q = 650J \quad W = 450J \quad \Delta E = ?$

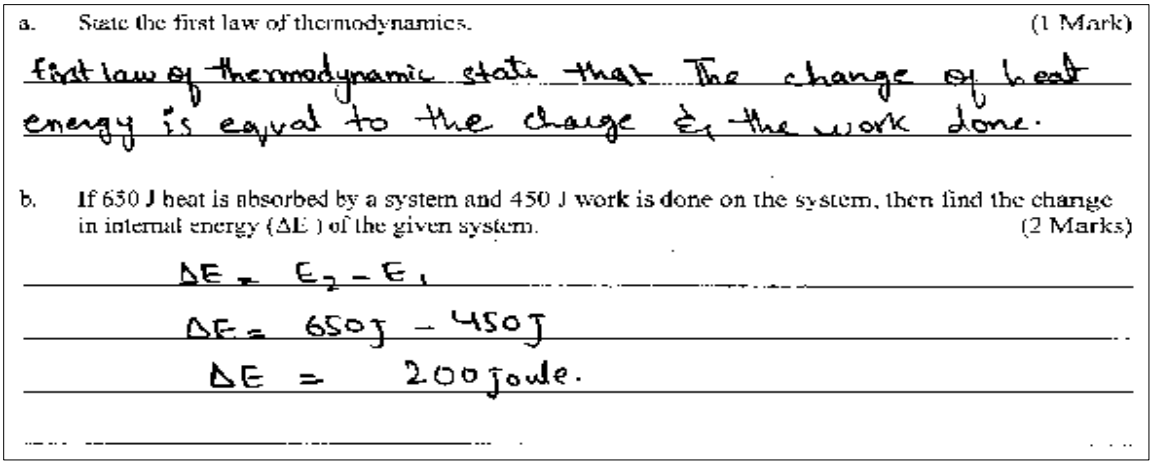
$\Delta E = q + w$

$\Delta E = 650J + 450J$

$\Delta E = 1100\ J$

Weaker responses wrote irrelevant or incomplete statement in relation to the first law of thermodynamics. Instead of giving the accurate statement for the first law of thermodynamics these response discussed about enthalpy, entropy, heat absorbed or released and work done etc. In part b, candidates were unable to calculate the change in internal energy of the given system correctly. These responses used incorrect formula or substitution of values and calculated wrong answer with no unit. Few of these responses used appropriate formula but were unable to calculate correct answer due to incorrect substitution of values.

Example:


a. State the first law of thermodynamics. (1 Mark)

first law of thermodynamic state that The change of heat energy is equal to the change of the work done.

b. If 650 J heat is absorbed by a system and 450 J work is done on the system, then find the change in internal energy (ΔE) of the given system. (2 Marks)

$\Delta E = E_2 - E_1$

$\Delta E = 650J - 450J$

$\Delta E = 200\ joule.$

Extended Response Questions (ERQs)

These questions offered a choice between part **a** and **b**

Most of the candidates attempted **12b** and performed better compared to **12a**.

Question 12a:

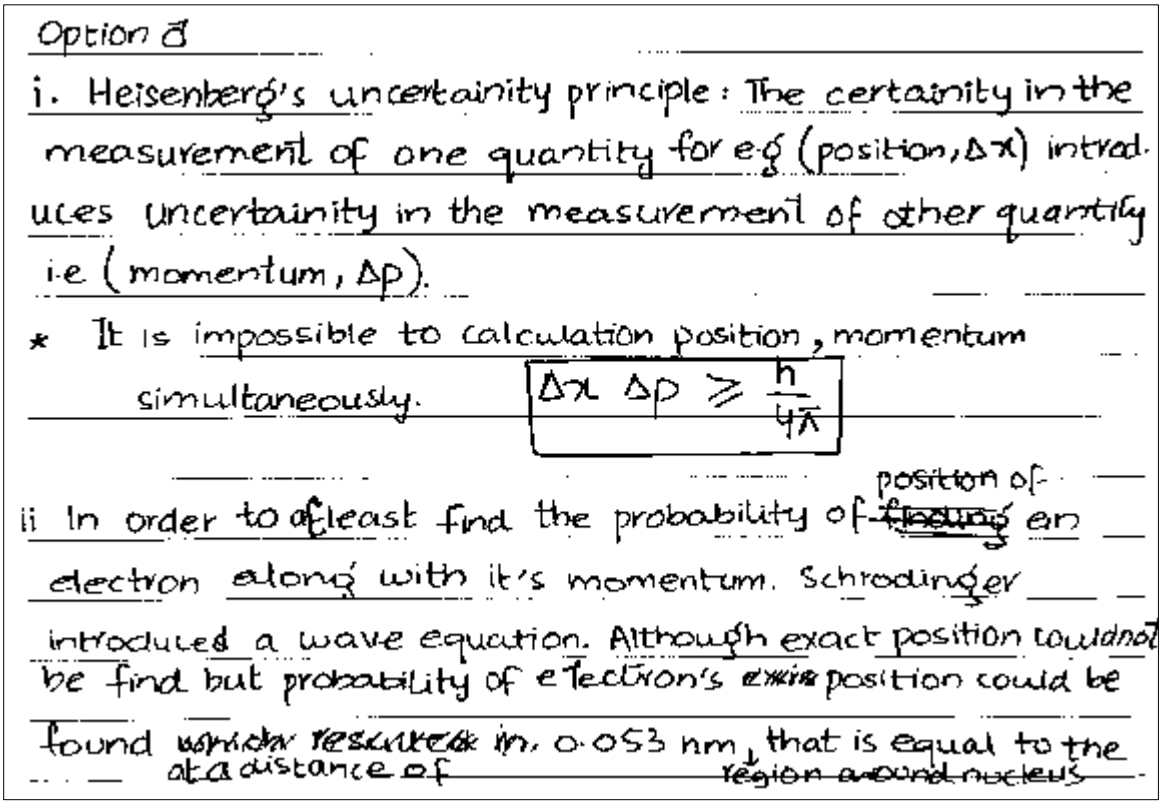
i. State Heisenberg's uncertainty principle and give its mathematical expression.

ii. Describe the concept of orbital on the basis of Heisenberg's uncertainty principle.

iii. Write any THREE differences between orbit and orbital.

Better responses clearly stated Heisenberg's uncertainty principle and gave its mathematical expression. Candidates mentioned that it is not possible to measure simultaneously the exact position and velocity of electron. In addition to correct mathematical expression, they identified 'Δx' as the uncertainty in the measurement of the position and 'Δp' as the uncertainty in the measurement of momentum of an electron. In part ii, they gave correct explanation about the orbital using Heisenberg's uncertainty principle and mentioned that the maximum probability of finding the electron is at a distance of 0.053 nm from the nucleus. In part iii, these responses provided correct differences between orbit and orbital in terms of their existence in an atom, motion of electron around nucleus, capacity of electron and their shapes.

Example:

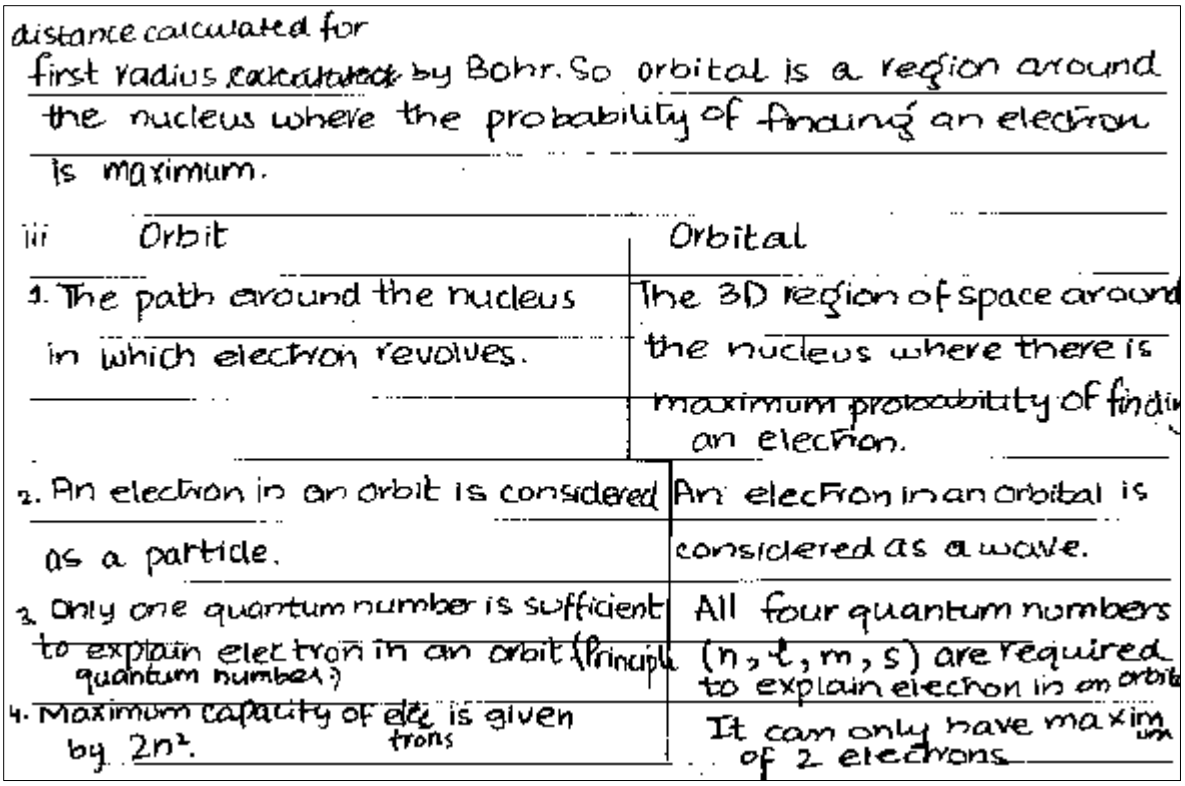


Option d

i. Heisenberg's uncertainity principle: The certainity in the measurement of one quantity for e.g (position, Δx) introduces uncertainity in the measurement of other quantity i.e (momentum, Δp).

* It is impossible to calculation position, momentum simultaneously. $\Delta x \, \Delta p \geq \frac{h}{4\pi}$

ii In order to atleast find the probability of ~~finding~~ position of an electron along with it's momentum. Schrodinger introduced a wave equation. Although exact position couldnot be find but probability of electron's ~~exist~~ position could be found ~~where~~ at a distance of ~~resulted~~ in. 0.053 nm, that is equal to the region around nucleus

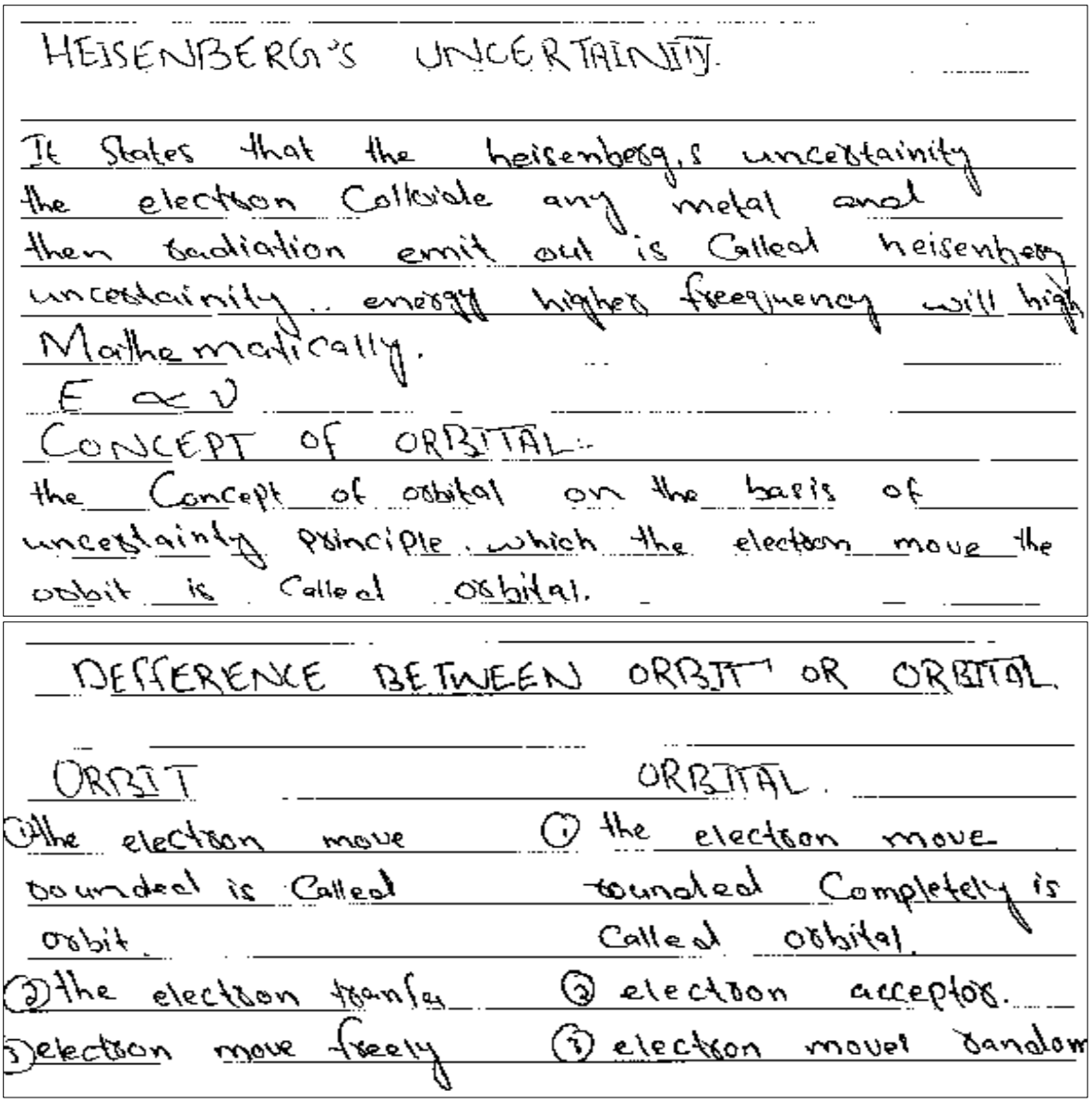


distance calculated for
first radius calculated by Bohr. So orbital is a region around the nucleus where the probability of finding an electron is maximum.

iii	Orbit	Orbital
1.	The path around the nucleus in which electron revolves.	The 3D region of space around the nucleus where there is maximum probability of finding an electron.
2.	An electron in an orbit is considered as a particle.	An electron in an orbital is considered as a wave.
3.	Only one quantum number is sufficient to explain electron in an orbit (Principle quantum number)	All four quantum numbers (n, l, m, s) are required to explain electron in an orbital
4.	Maximum capacity of electrons is given by $2n^2$.	It can only have maximum of 2 electrons

Weaker responses exhibited that candidates were unable to explain the Heisenberg's uncertainty principle and its mathematical expression. They gave illogical statement and irrelevant details in relation to the Heisenberg's uncertainty principle. Some candidates attempted to discuss the uncertainty in position or uncertainty in momentum, but did not give complete and accurate statement with mathematical expression or symbols such as Δx or Δp. Few of these responses stated the Heisenberg's uncertainty principle correctly, but showed wrong mathematical expression. In part ii, candidates were unable to describe the concept of orbital in the light of Heisenberg's uncertainty principle. They did not provide any relevant information such as the volume of space in which there is 95% chances/ maximum probability of finding an electron or the distance of 0.053 nm from the nucleus. Most of them tried to explain orbital using generic statement like orbitals are the places around nucleus where electrons are present. In effort to describe the concept of orbital, some of these responses gave definition of orbits. In part iii, candidates did not differentiate between orbit and orbital correctly. They had no information regarding the shape or types of orbitals and the maximum capacity/ number of electrons that each orbital or orbit can accommodate. However, few of these responses exhibited differentiation between orbit and orbital in terms of two dimensional and three dimensional motions of electrons around the nucleus.

Example:



HEISENBERG'S UNCERTAINTY.

It States that the heisenberg's uncertainity the electron Collside any metal anod then radiation emit out is Called heisenberg uncertainity.. energy higher freequency will high Mathematically.

$E \propto \nu$

CONCEPT OF ORBITAL:-

the Concept of orbital on the baris of uncertainty Principle. which the electron move the orbit is Called orbital.

DEFFERENCE BETWEEN ORBIT OR ORBITAL.

ORBIT	ORBITAL.
(1) the electron move bounded is Called orbit.	(1) the electron move bounded Completely is Called orbital.
(2) the electron transfer	(2) electron acceptor.
(3) electron move freely	(3) electron moved random

Question 12b

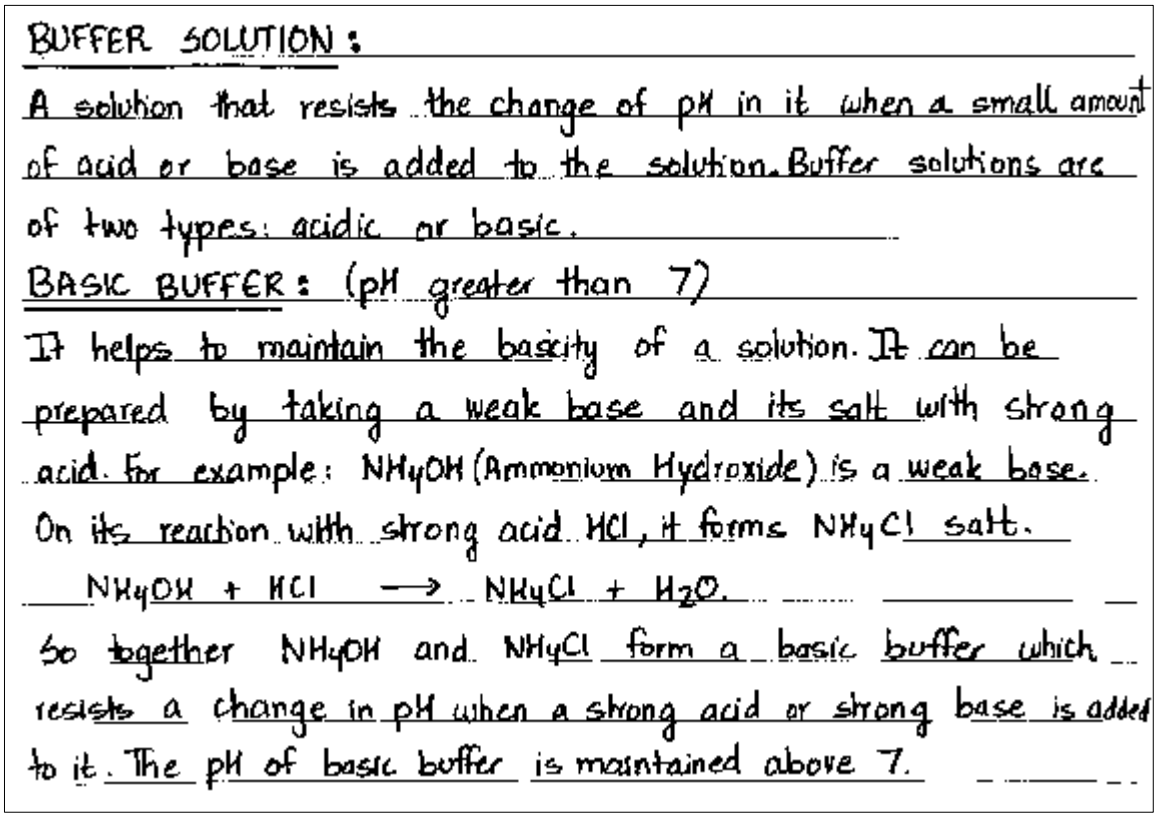
i. Define buffer solution.

ii. Describe the preparation of a basic buffer. Give ONE example of it.

iii. An acidic buffer composed of CH_3COOH and CH_3COONa helps to maintain the pH of a solution when an acid or a base is added to it. Justify the given statement with the help of chemical equations.

Better responses provided correct description about the buffer solution. In part ii, candidates showed good understanding of the types of buffer and hence were able to identify basic buffer and describe its preparation. They mentioned that the basic buffer is prepared by mixing a weak base and a salt with a strong acid. Most of these responses considered ammonium chloride-ammonium hydroxide buffer solution as an example of basic buffer. In part iii, candidates inferred the given situation clearly and explained the effect of acid or base on acidic buffer composed of CH_3COOH and CH_3COONa. They described the action of buffer with the help of chemical equations under both given conditions.

Example:



BUFFER SOLUTION:

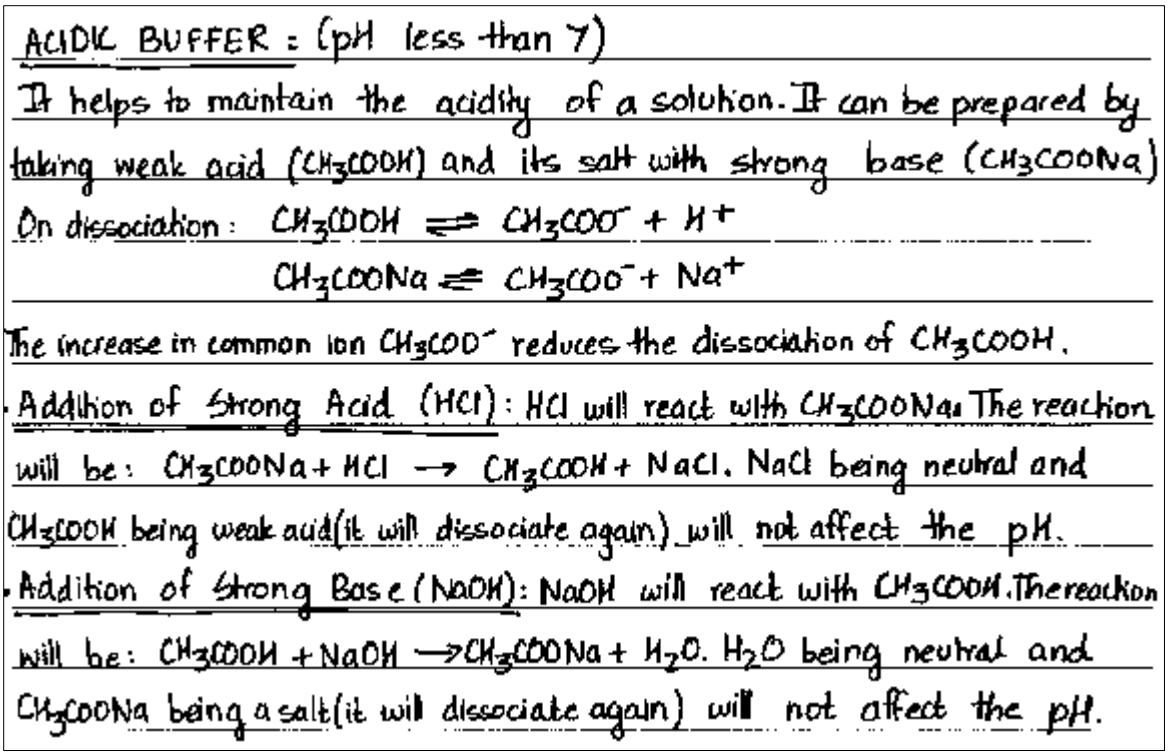
A solution that resists the change of pH in it when a small amount of acid or base is added to the solution. Buffer solutions are of two types: acidic or basic.

BASIC BUFFER: (pH greater than 7)

It helps to maintain the basicity of a solution. It can be prepared by taking a weak base and its salt with strong acid. For example: NH_4OH (Ammonium Hydroxide) is a weak base. On its reaction with strong acid HCl, it forms NH_4Cl salt.

$NH_4OH + HCl \longrightarrow NH_4Cl + H_2O$.

So together NH_4OH and NH_4Cl form a basic buffer which resists a change in pH when a strong acid or strong base is added to it. The pH of basic buffer is maintained above 7.



In most of the *weaker responses,* candidates were unable to provide correct definition of buffer solution. However, in a few cases, candidates were able to define buffer solution, but did not perform well in the next two sections of this question. In part ii, they showed lack of knowledge about the concept of buffer and were not able to identify basic buffer (as the buffer with pH more than 7) and provided completely wrong description for the preparation of basic buffer. These responses mentioned that the basic buffer is prepared by mixing a specific amount of strong base with acidic salt. Furthermore, these responses gave irrelevant examples of basic buffer such as NH_4Cl with HCl. In part iii, they did not analyse the given situation with acidic buffer correctly. These responses did not identify the given acidic buffer composed of CH_3COOH and CH_3COONa (as the buffer with pH less than 7) and provided illogical explanation in relation to the action of buffer with no chemical equation to show equilibrium direction under both conditions. The requirement of the question was a discussion ‘under acidic condition hydrogen ion combines with the acetate ion to make acetic acid, which is a weak acid or under basic condition hydroxide ion react with an acetic acid molecule to form acetate ion and water’. Instead of the aforementioned discussion, candidates wrote generic statement like ‘under acidic or basic conditions buffer maintain its pH because under these conditions it dissociates to form ions which exchange each other and help to maintain pH of solution’.

Example:

(i) Buffer Solution: A solution whose pH can not be further changed by addition of an acid or a base.

(ii) Preparation of Basic Buffer:

- Add two component .i.e. a ~~weak acid~~ strong base and a ~~basic~~ acidic salt.
- Add 60% water to maintain pH of solution
- Add more 40% water to maintain volume of solution.

Example: HCl + NH_4Cl

(iii) Buffer is a solution whose pH does not change by addition of an acid or a base Thus, the pH of a buffer is maintained when an acid or a base is added to it. It is because ions of buffer and ~~salt~~ acid/base dissociates and ~~the~~ exchange eachother's ions. And this process helps buffer to maintain its pH when an acid or base is added to it.

Most of the candidate attempted **13b** and performed better compared to part **13a**.

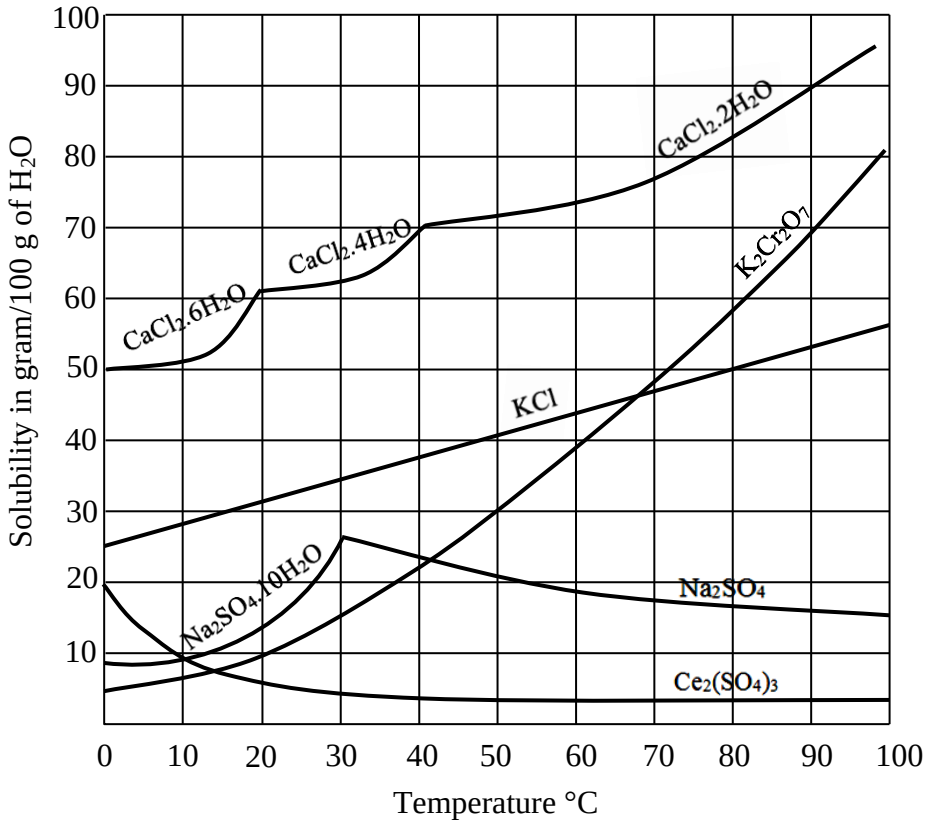
Question 13a:

i. Define solubility. Using the given graph, describe in detail the types of solubility curves shown for each salt.

ii. Deduce the following from the graph.

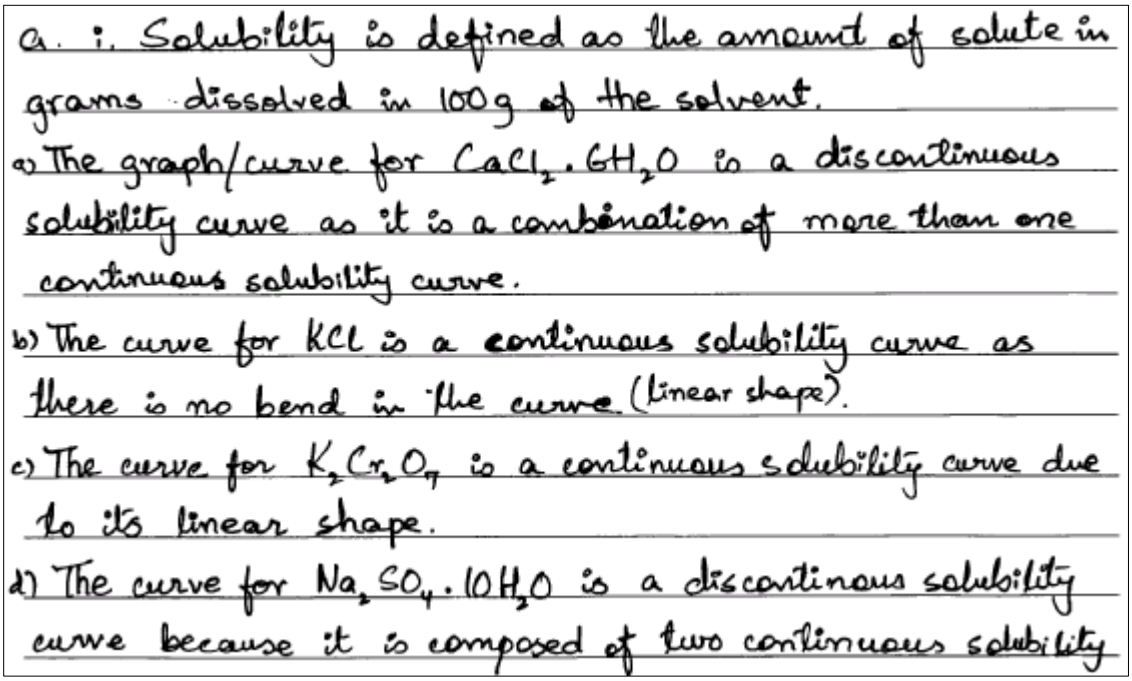
I. Grams of undissolved potassium dichromate that settles at the bottom of the beaker when 70 g of it is mixed in 100 g of water at a temperature of 50ºC.

II. Type of potassium chloride solution as saturated or unsaturated or supersaturated when 50 g of it is dissolved in 100 g of water at 80°C.



Better responses correctly defined solubility and identified each type of solubility curves given for each salt. These responses recognised solubility curves of KCl, $K_2Cr_2O_7$ and $Ce_2(SO_4)_3$ as continuous curves. Furthermore, they discussed about the increase or decrease of solubility or temperature in appropriate curve. They identified solubility curves associated with $Na_2SO_4.10H_2O$ and $CaCl_2.6H_2O$ as discontinuous curves. Moreover, they discussed about water of crystallization and phase changes. In part ii, they calculated correct amount of undissolved potassium dichromate under given condition, and identified the type of potassium chloride solution as saturated.

Example:



a. i. Solubility is defined as the amount of solute in grams dissolved in 100g of the solvent.

a) The graph/curve for $CaCl_2.6H_2O$ is a discontinuous solubility curve as it is a combination of more than one continuous solubility curve.

b) The curve for KCl is a continuous solubility curve as there is no bend in the curve (linear shape).

c) The curve for $K_2Cr_2O_7$ is a continuous solubility curve due to its linear shape.

d) The curve for $Na_2SO_4.10H_2O$ is a discontinous solubility curve because it is composed of two continuous solubility curves.

e) The curve for $Ce(SO_4)_3$ is a continuous solubility curve as it doesn't contain any bend.

ii. I. Solubility of $K_2Cr_2O_7$ at 50°C is 30g per 100g of H_2O.

70g of $K_2Cr_2O_7$ makes a supersaturated solution.

So, mass of undissolved $K_2Cr_2O_7$ = 70g − 30g

= 40g

II. Solubility of KCl at 80°C is 50g per 100g of H_2O.

So, when 50g of KCl is dissolved in 100g of H_2O at 80°C, the solution is saturated.

Weaker responses were unable to correctly define the term solubility. Candidates did not write about the ability of substance to dissolve in a solvent or the maximum amount of solute dissolved in a solvent at equilibrium. Furthermore, these responses discussed the trend of solubility for each salt at different temperature, but were not able to identify the types of solubility curves given for each salt. For example, candidates read the given solubility curve of $K_2Cr_2O_7$ and mentioned that the solubility of $K_2Cr_2O_7$ is 10 g/ 100 g of H_2O at 20°C, and it increases at 50°C and maximum at 100°C, but they did not identify it as continuous or discontinuous solubility curve. Candidates provided similar information for other solubility curves at different temperature. Additionally, these responses did not provide discussion in relation to water of crystallization and phase changes occurred in solubility curves of $Na_2SO_4.10H_2O$ and $CaCl_2.6H_2O$. In part ii, they were not able to interpret the given situation clearly and provided wrong calculation to calculate the amount of undissolved potassium dichromate. Some of these responses used the solubility curve of KCl instead of $K_2Cr_2O_7$ and calculated wrong answer as 30 g. Finally, these responses further identified the type of potassium chloride solution incorrectly as super saturated instead of identifying it as saturated.

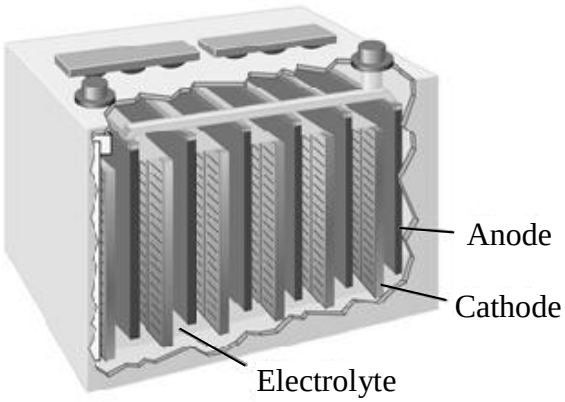
Example:


Solubility :- Amount of Solute dissolved in Specific volume of Solvent.
-> $k_2Cr_2O_7$ = It Solubility at 20°C is 10g/100g while at 50c it reaches at 40g/100g and at 100°C its solubility is 80g/100g. -> $Na_2SO_4.10H_2O$ = It Solubility at 30°C is around 25g/100g and then it decrees with time -> $Ce_2(SO_4)_3$ = it Solubity at 0°C is 20g/100g and then it decreases as increase in temprature.
-> KCl = Its Solubity at 0°C is around 25g/100g as temprature is increasing

its Solubility is increasing and at 100°C its Solubility is around 55g/100g. -> $CaCl_2.6H_2O$ = Its Solubility at 0°C is 50g/100g. at 20°C it reaches at 60g/100g. At 40°C it reaches at 70g/100g. and then it increase with increase in temperature.
B) i). Grams of Undissolved at 50°C are 40g so => 70 - 40 = 30 g of Un dissolved Potassium dichromate.
ii). Grams at 80°C are 50g
so 50 - 50 = 0.
So at 80°C Potassium Chloride is Super Saturated.

Question 13b:

The given diagram shows a lead storage battery.



i. Describe the lead storage battery with reference to the following.

   I. Name of anode, cathode and electrolyte
   II. Number of cells connected in series to generate 12 volts
   III. The most common uses of lead storage battery in daily life

ii. Write balanced equations for the following chemical reactions that occur during the discharge of a lead storage battery.

   I. At the cathode
   II. At the anode
   III. Overall cell reaction

Better responses described the lead storage battery by correctly identifying the metallic lead (Pb) as anode, lead oxide (PbO_2) as cathode and H_2SO_4 as electrolyte present. Candidates mentioned that as 1 cell produces 2 volts, so 6 cells are required to generate 12 volts. These responses mentioned that the use of lead storage battery in car is the most common. In part ii, they showed a balanced chemical equation for the chemical reactions that occur during the discharge of a lead storage battery.

Example:

LEAD STORAGE BATTERY/LEAD ACCUMULATOR:

It is a type of galvanic/voltaic cell that can produce electrical energy from a non-spontaneous chemical reaction inside the cell. It is secondary storage battery and rechargeable. It is commonly used as a car battery in real life, to start the engine of a car or to switch on its lights. It produces electricity by discharge. The anode is spongy lead (Pb) while the cathode is powered lead oxide PbO_2. The electrolyte is 30% H_2SO_4 (density = 1.25 g/cm³). 1 lead storage cell generally produces 2V so to produce 12 volts, six lead battery cells are connected together in series. The following reactions take place inside the battery:

DISCHARGING:

- ANODE (Pb)

The lead (Pb) combines with SO_4^{-2} ions from electrolyte to form $PbSO_4$ which deposits at the anode. 2 electrons are released.

$Pb_{(s)} + SO_{4(aq)}^{-2} \longrightarrow PbSO_{4(s)} + 2e^-$ (Oxidation)

These $2e^-$ move towards cathode so electricity is produced.

- CATHODE (PbO_2)

PbO_2 is reduced by $2e^-$, $4H^+$ and SO_4^{-2} from electrolyte also participate in this:

$PbO_{2(s)} + 2e^- + SO_{4(aq)}^{-2} + 4H^+ \longrightarrow PbSO_{4(s)} + 2H_2O$ (Reduction)

- Overall Cell reaction is as follows:

$Pb_{(s)} + PbO_{2(s)} + 2SO_4^{-2} + 4H^+ \longrightarrow 2PbSO_{4(s)} + 2H_2O$

2 moles of $PbSO_4$ are produced in 1 cell, after its production discharge stops. The density of H_2SO_4 falls to 1.15g/cm³. It comes back to its original value during recharging. After recharging, cell can again produce $PbSO_4$.

Weaker responses were unable to provide required information in relation to the lead storage battery. Candidates wrongly identified metallic lead (Pb) as cathode and lead oxide (PbO_2) as anode. Some of these responses identified platinum as anode and silver as cathode and $PbSO_4$ as electrolyte instead of H_2SO_4. These responses mentioned that 4 cells in the lead storage battery produce 3 volts, so 4 cells are required be connected in series to produce 12 volts. Instead of mentioning the most common use of lead storage battery in daily life, these responses tried to discuss the function or working of lead storage battery. Some of these responses mentioned that the lead storage battery has wide application in different fields of life, but did not specify the equipment and where it is used exactly. In part ii, most candidates showed poor knowledge of the concept and did not mention the correct balanced chemical equation for the chemical reactions that occur at the cathode and anode including the overall cell reaction during the discharge of a lead storage battery. Some of these responses displayed incorrect and irrelevant equations with illogical loss/ gain of electrons, reduction/ oxidation and change of oxidation number without mentioning the electrolyte, so they were not able to write the overall reaction of the lead storage battery.

Example:

